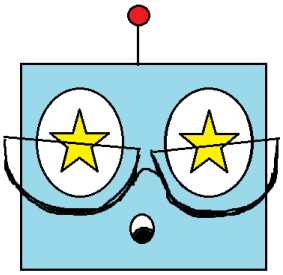
Design Project Report



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Human Centered Design

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Appendix **3**
Introduction **4**
Need Finding **5**
Personas 5
Process 7
Focus group questions 8
Focus group data 8
Interpretation 12
Design Goals **13**
List of Goals 13
User Board and Persona Journey 15
Design Description **17**
Design Concept 17
Detailed Design 21
Evaluation **33**
Wizard of Oz 33
Experimental Usability Study 35
Modifications 38
Summary **39**

Appendix

Figure 1: Craig Persona 5
Figure 2: Mia Persona 6
Figure 3: Harry Persona 7
Figure 4: Focus Group Data 8
Figure 5: Michelle S. Group 10
Figure 6: List of Goals 13
Figure 7: Storyboard 15
Figure 8: Mia User Journey 17
Figure 9: Log History Wireframe 18
Figure 10: Personalized Experience Wireframe 19
Figure 11: Create a New Chat Wireframe 19
Figure 12: Rate Your Chat Wireframe 20
Figure 13: Report a Bug Wireframe 20
Figure 14: Copy Link Wireframe 21
Figure 15: Home Page 22
Figure 16: Personalization Page 22
Figure 17: New Chat Page 23
Figure 18: User Help Page 25
Figure 19: Chat Rating Page 26
Figure 20: Filled Out Chat Rating Page 26
Figure 21: Feedback Confirmation Page 26
Figure 22: Chat Log Page 28
Figure 23: Saved Chat 28
Figure 24: Share Page 29
Figure 25: Report a Bug Page 31
Figure 26: Report Bug Confirmation Page 32
Figure 27: Experiment Results Table 36
Figure 28: Experiment Results Graph 37

Introduction

The overall purpose of the app we have designed is to aid people with social anxiety through a friendly chat, help professionals find inspiration by brainstorming, and provide a personalized experience for each user through the program's memory. This design responds to the requirement that the user interacts with AI by providing a tool that allows the user to input text and receive responses from an AI through a chatbot-like design. Some similar apps that we used for inspiration are iMessage, with their use of text bubbles and different colored responses along with the dictation feature to record your voice and send that in as input, and Facebook Messenger, for their 16:9 web-based interface. We also took inspiration from chatGPT because it is the forefront program in the category we are designing for, and they have done many things right. For example, users can rate responses from the chatbot. The chatbot can remember previous conversations through a history feature and tailor responses to you through a private account. Our main goal is to help people with social anxiety, but we want our design to be versatile for any use case a chatbot fits.

Need Finding

Personas

Figure 1: Craig Persona

User 1: Craig is married and has three children. He likes to go on camping trips with them and make smores. Craig follows a strict schedule of exercising five times a week in the mornings and going to bed early. He loves to explore new technologies, especially when it comes to improving his work. Craig goes the extra mile at his workplace for two reasons. The first is that he is passionate about marketing. The second is that Craig wants to show his suitability for the job and hopes to get promoted. The only problem is that he is currently at a creative block. He hopes to find a new technology to help get over this road bump. See Figure 1 above for

details on Craig.

Figure 2: Mia Persona

User 2: Mia is a daughter of two parents and a sister to one sibling. She is currently single. Mia enjoys hot chocolate in the winter and prefers tea over coffee. She spends most of her time with the same few friends and attending class. Mia is a university student pursuing a degree in English. Her dream is to become an editor for the local newspaper. Though she wishes to achieve her dream job, she is worried that her social anxiety will come to prevent her from doing so. She hopes to find a new technology to help her overcome her challenges. See Figure 2 above for details on Mia.

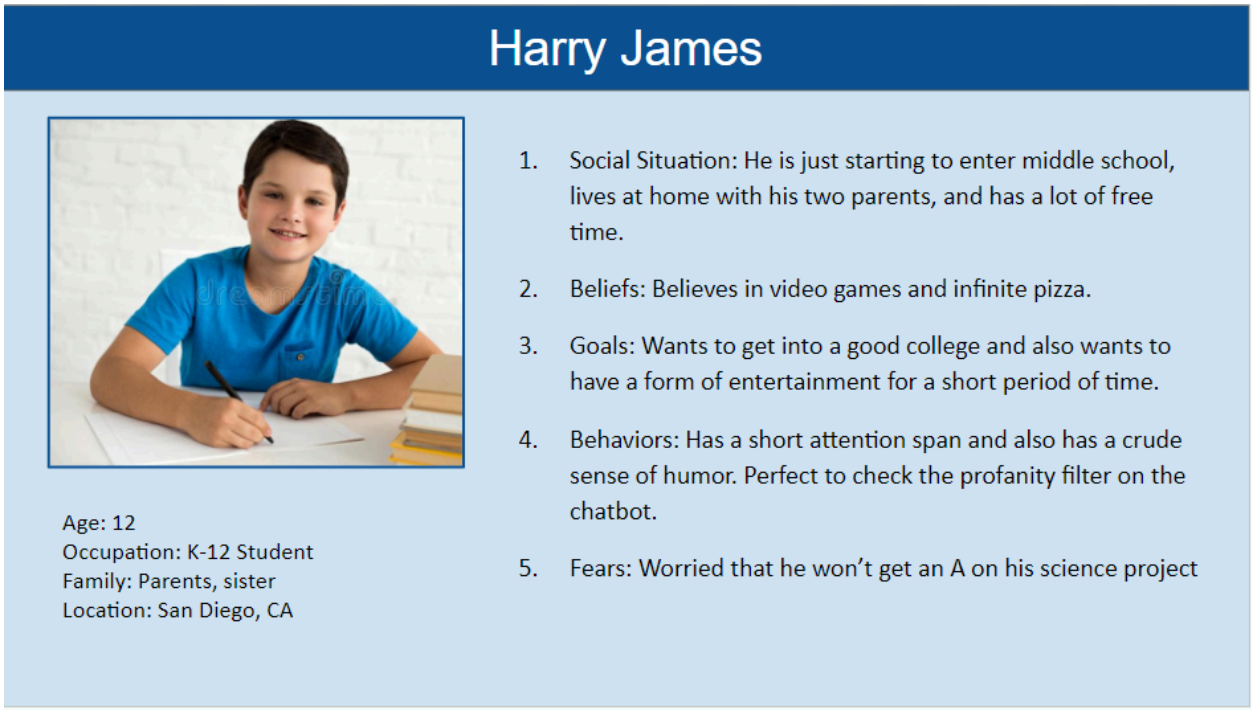


Figure 3: Harry Persona

User 3: Harry is a son of two parents and a brother of one sibling. He loves playing video games like Fortnite and pepperoni pizza. Harry is starting sixth grade and is excited about this new chapter. He dreams of getting into a good university and wants top-notch grades. The only thing is Harry is worried that he won't get an A on his science project. Harry hopes to find a new technology to help him bounce off ideas and improve. See Figure 3 above for details on Harry.

Process

While setting up the interviews, the best possible participants were people of all ages, with a preference for people under 65. The questions below were given to the participants during the interview. These questions prompt the participants to provide feedback on the product and what can be done to improve it. The focus group was held with young adults, both students and professionals, to get unique angles on how the product should perform. The following displays the interviewees' responses to the questions and feedback on the product.

Focus group questions

1. What chat app (iMessage, Instagram, facebook messenger, etc) do you use the most and why?
2. What are some concerns you may have about talking to a chatbot about personal information?
3. Would you be interested in using this type of app? If not, do you know someone who would?
4. Do you have any ideas on how we can make our ideas better in the design phase?
5. Do you feel this concept would work the best as a chatbot or would it work better in a different format such as a voice assistant/video conference?
6. Do you have any critiques for me and the concept?

Focus group data

	What chat app do you use the most and why?	**What are some concerns you may have about talking to a chatbot about personal information ?**	**Would you be interested in using this type of app? If not, do you know someone who would?**	**Do you have any ideas on how we can make our ideas better in the design phase?**	**Do you feel this concept would work best as a chatbot or would it work better in a different format such as a voice assistant/video conference ?**	**Do you have any critiques for me and the concept?**

Sean C Group:	The chat apps they use were Whatsapp, iMessage, and Facebook Messenger . The reasons for using these were because iMessage is the default one on iPhones, Facebook Messenger allows them to communic ate with people who you only have on Facebook and whats app has good encryption .	Some concerns the people I talked to had about personal information were that their personal information would be shared or viewable by others, or having the possibility of the chatbot being hacked.	Some said they would be interested in the chatbot if it was useful and had good security and privacy measures to be sure data wouldn't leak, Some said they themselves don't see a need for it in their lives but can imagine the target audience would benefit from using it	Some suggestion s based on the design were allowed to choose how the chatbot responds possibly by allowing the user to choose the personality they wish, giving the user a way to report bugs or flag issues, as well as giving it a rating system to rate how useful the results were so that the chatbot can be trained while it's being used.	Most said that the chatbot would work the best because you can see what's going on as you interact with it but some said that they would find a voice assistant version to be useful as well.	Some critiques and suggestion s were to make the chatbot work in multiple languages as well as learn from the user over time to have more personalize d messages. Another suggestion was to have it integrate with other programs like the calendar or reminder app on your phone.
Amanuel H Group	They used iMessage and Snapchat for their messaging apps.	The potential for advice given back to the user to not be 100% correct	They said that they would potentially be interested in a product like this.	A warning should be added disclosing that the advice given by the AI may not be	They believed that the chatbot would be the most efficient way to do this idea.	Advice should be kept at a minimum to reduce the chances of users making big

				accurate		decisions based off of the answers of the AI
Mena T Group	They use Instagram and snapchat along with iMessage as well.	One participant was worried about hacking and if the information could be shared.	When asking them if they would use this, one participant said they would but the other wouldn't, unless for practice interviews, etc.	They thought that the design should be friendly and welcoming since it is going to act as a virtual companion	The chatbot would be the best way to implement this type of design.	If it would learn about the user over time to make it more personable, or if it would just reset. If it were to reset, they came to the decision that it would not help with social anxiety too much.

Figure 4: Focus Group Data

Michelle S Group	Responses
What apps do you use for communication on a weekly basis?	All of the group uses discord to communicate on a daily basis, with instagram, snapchat, and basic text messaging being used occasionally. Discord is used for general communication and communication during other activities such as playing games and watching shows in a group setting, while instagram is used for sending video or photo content to others. Personal messages are sent through snapchat and text.
Do any of you experience social anxiety?	The majority stated that they have experienced social anxiety before and continue to experience it daily. One group member did not experience social anxiety.

If you have, have you looked to digital simulations to improve your social skills?	All four responded that they have not.
Have you ever used a chatbot before?	All of the participants have used a chatbot before, specifically ChatGPT.
If you have used a chatbot, why did you use it and how long?	One of the participants has used a chatbot once for programming information. It was used for two days total and for a personal programming project. Two of the other group members used it for about 30 minutes just to see what it was about. One tried using it to solve complex algebra equations for a simulation, essentially using it as an advanced calculator. They stated they tried this for about 45 min.
If you have used a chatbot before, how did you discover it?	The participants listed social media (specifically reddit r/programminghumor) and word of mouth as their sources.
Would you ever use a text-to-speech option in a communication application?	All of the participants stated that they have and would, but only in a joking manner.
Would you ever use a speech-to-text option in a communication application?	All of the participants have said yes, with multiple having used it before to write essays and messages to other people.
Would you use it with a chatbot/ would you speak to the chatbot?	Two participants stated that they would, while two said they would not. Those who were willing to speak to the chatbot stated that "google assistant already does it and so does siri", while those who would not stated that they would be uncomfortable.
Would you feel uncomfortable communicating with an AI with a face/animated model?	All of the participants stated that they would not feel uncomfortable, however they would not seek out an AI or chatbot because of the model.
Do you believe that a chatbot can assist those with social anxiety?	All participants agreed that it would help those with social anxiety, but to varying effects depending on the user.

Figure 5: Michelle S Group

Interpretation

After reviewing the Focus Group data, we collected essential information from the many questions we created and asked. We decided to have more freedom with our questions, as not all of us who ran focus groups asked the same questions. This way, we have more information and data to build our design. We learned from our focus groups that users want the design to be friendly and transparent. Since the product will aid people with social anxiety, it shouldn't be visually striking or challenging. Many participants also had concerns about the AI aspect of the end design. There were privacy concerns when dealing with potentially sensitive information and the question of the accuracy of the chatbot itself. We also learned that despite these privacy concerns, many participants believed the chatbot would be more valuable if it collected data and learned from each conversation. That way, the chatbot would become your personalized assistant or friend to talk to because it remembers things from your previous discussions. With this information, we decided to have our chatbot remember conversations and implement a rating system. The user can rate a response given by the chatbot as a thumbs up or thumbs down, indicating the usefulness of the answer. There is also the option to rate the conversation and report any issues to the development team.

Design Goals

The Design goal for our app is to create a chatbot that responds to user input to satisfy their needs. The application should be easy for people of all ages and assist users in any way. To help with ease of use, functions like tech support will be implemented. The chatbot will be designed for entertainment, professional help, and social interaction. Our list of design goals is listed as follows:

- Implementing typing and voice recognition input
- History of saved chat logs that can be accessed and also erased
- User's ability to give feedback on their experience through tech support and/or rating
- Learning capabilities that allows that chat bot to learn new information through interactions with users
- Chatbot's ability to give accurate responses to any topic available

The look of the chatbot is going to be very simplistic. We don't want the actual webpage to be very cluttered. Most of the webpage is going to be used for actual interaction with the chatbot.

List of Goals

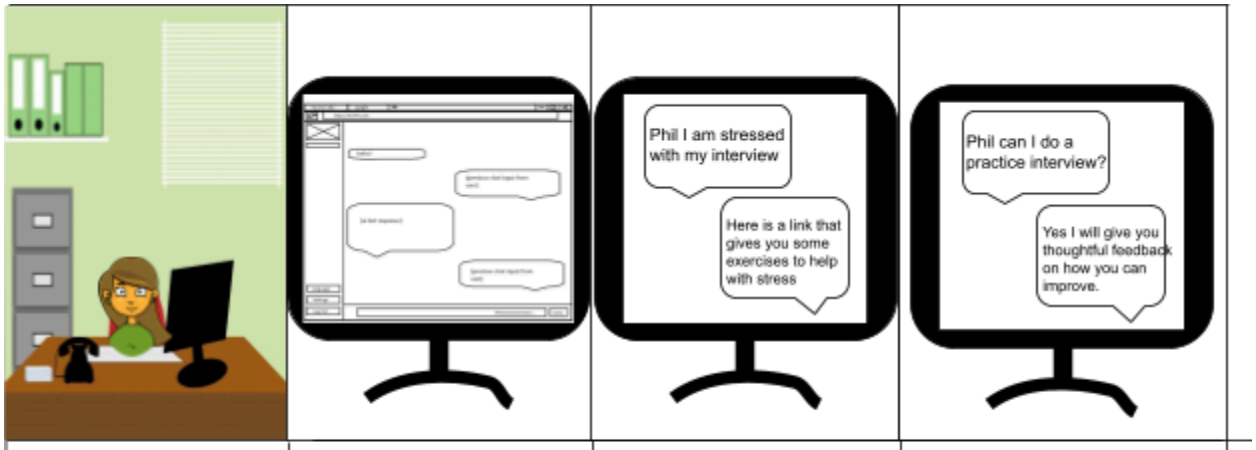
Goals	Usability Design Principle	App Purpose
Chatlog History	Usefulness	Users should be able to access old conversations that they had with the chat log. Users also should be able to erase chats as well.
Feedback	Efficiency of Use	Allows users to contact tech

		support if there is an issue. The user must be able to give feedback and ratings to the creators of the chatbot.
Accurate Responses	Usefulness	The chatbot must be able to give responses of those like a human. It must give responses on any subject, like educational, personal, etc.
Typed input and voice recognition input	Efficiency of Use	Users should have the option of either physically typing in their input or using voice recognition for the user's input. This will help with ease of use.

Figure 6: List of Goals

We have not altered most of our goals because they align with what the focus groups are asking for, a “friendly and welcoming” companion. We have decided to add the design goal of user privacy due to the concerns raised during the focus groups. Next, we added the goal of making the experience “personable” to the user. Lastly, there was outstanding feedback in regards to the chatbots variety and intelligence in a subject. Therefore, we have decided to add the goal of having the chatbot give the most accurate responses possible.

User Board and Persona Journey



Mia has an interview coming up. She is currently in her at home office thinking of ways to prepare for her upcoming interview.	Mia decides to use Phil the Chatbot. She logins in to the webpage where she is able to communicate with the chat bot. She also sees how there are settings that she can adjust as needed	The first thing that she types into the chatbot is that she is stressed for her interview. Phil then responds by sending her a link of possible exercises that can reduce stress.	She then asks the chatbot if she can do a practice interview. Phil responds that he can and will provides her with thoughtful feedback and possible methods to stand out in her interview

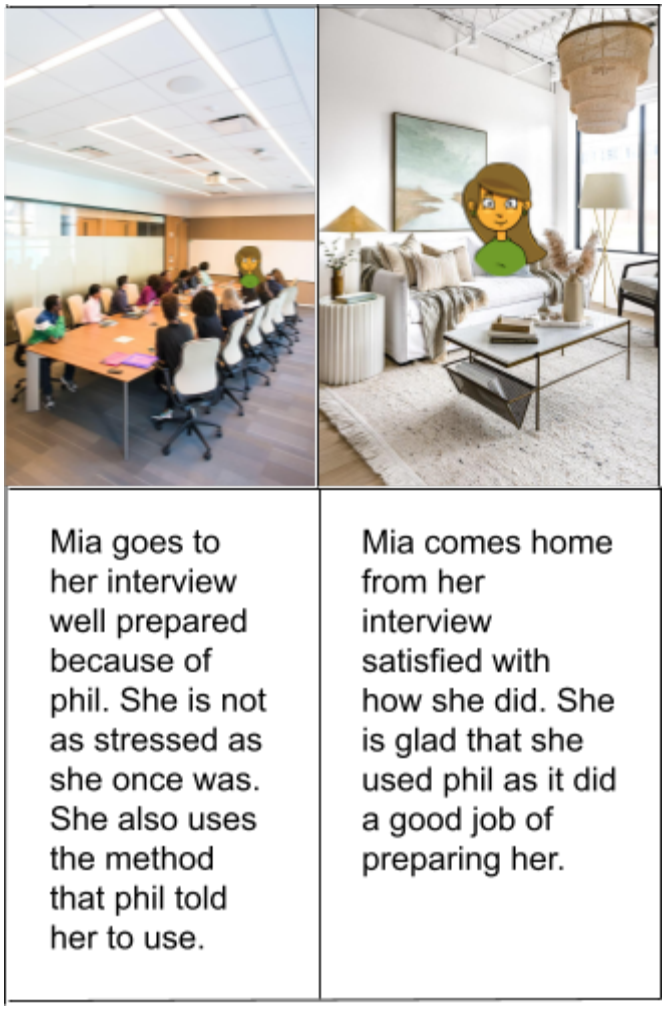


Figure 7: StoryBoard

Design Description

Design Concept

Low-Fidelity Prototypes:

When it comes to Low Fidelity Prototypes, two steps are taken. One is the user journey which is a detailed description of a concept audience member who would be interested in the product or someone who would benefit from it overall. This description will include their current situation that needs a solution, the steps they take to use the products to solve their issue, and their current state of being (emotions). We can take one of the personas previously mentioned, Mia, to showcase what a user journey would look like:

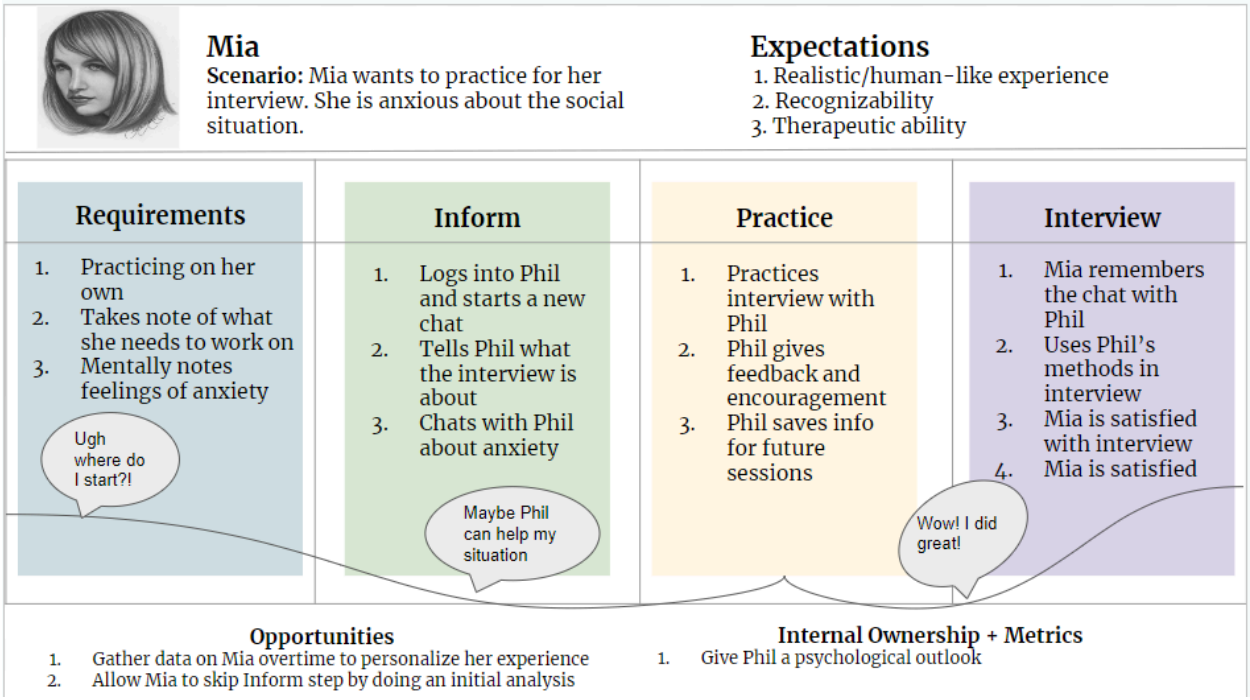


Figure 8: Mia User Journey

It takes ambition to create something that can help others, but one of the complex parts is designing something practical. This is where the second step comes in, the wireframe. The

wireframe helps create a vision for what the product might look like. This is done using either online tools or just pencil and paper. Making a rough sketch for your future product will help you find things to include in your final design to help your target demographic with their issues.

Here are the wireframes we made to help us with brainstorming and implementing new ideas into our future Chat Bot:

1. User checks their chat log:

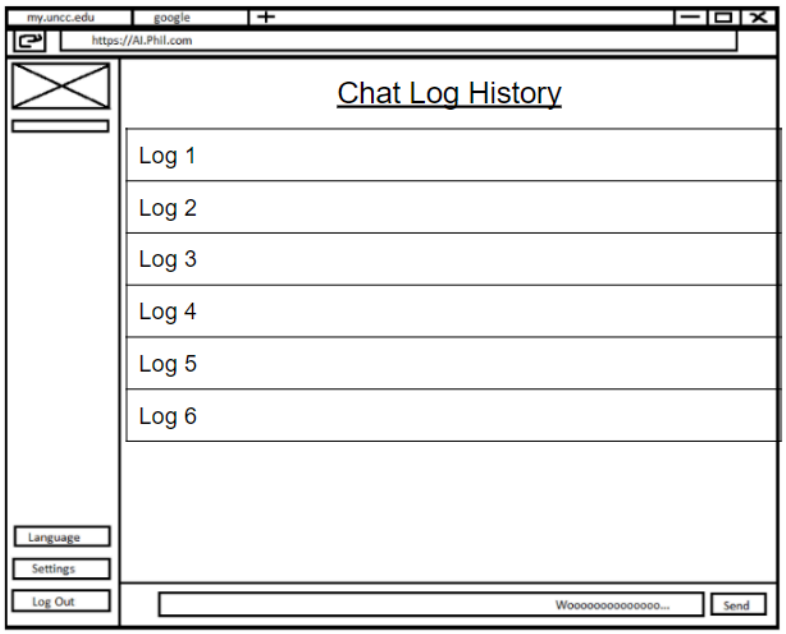


Figure 9: Log History Wireframe

2. User takes a survey for a personalized experience:

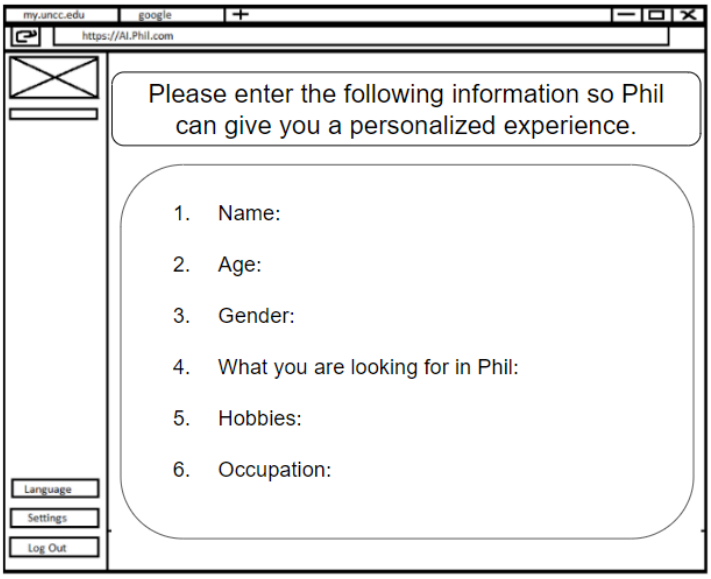


Figure 10: Personalized Experience Wireframe

3. User Chats with the Bot:

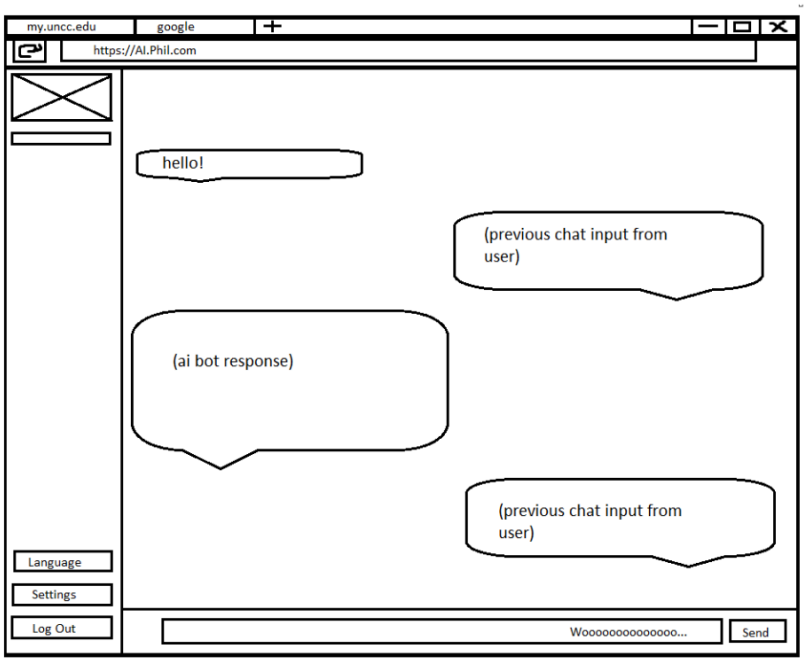


Figure 11: Create a New Chat Wireframe

4. User rates their chat with the bot:

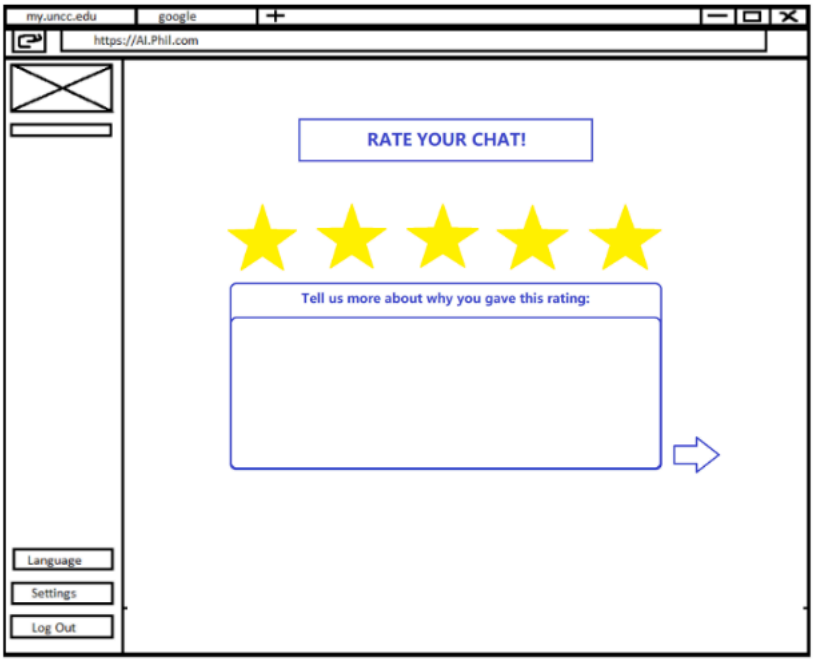


Figure 12: Rate Your Chat Wireframe

5. User reports any bugs they saw:

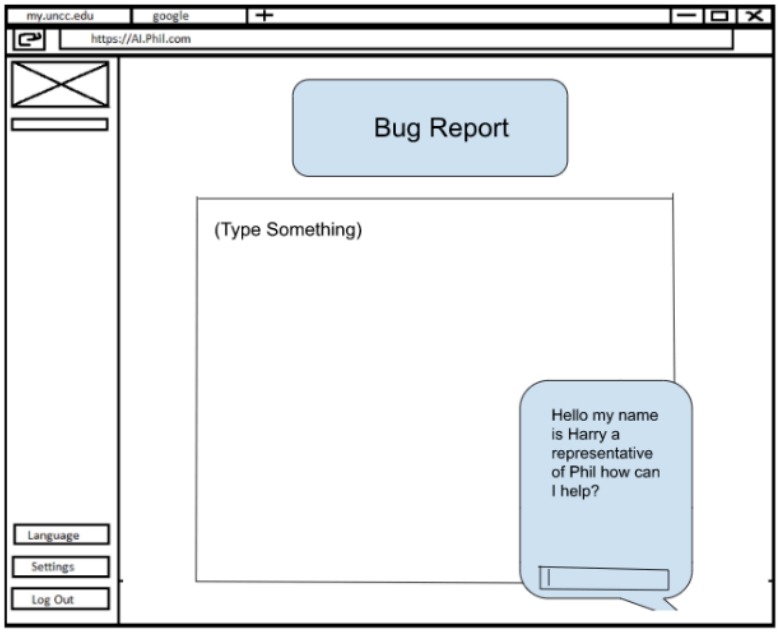


Figure 13: Report a Bug Wireframe

6. User accesses the share button

Figure 14: Copy Link Wireframe

Detailed Design

High-Fidelity Prototype:

We have completed our initial brainstorming session and found most things we can implement to further assist our user base, so we made a High-Fidelity Prototype. High-Fidelity Prototypes differ from Low-Fidelity ones in a few ways and are some of them:

- High-Fidelity Prototypes include interactable sections/targets whereas Low-Fidelity ones do not.
- There are clickable links within a High Fidelity Prototype whereas Low-Fidelity Prototypes include screens showing what the links would take them to without the benefit of being interactable.
- Fonts, Graphics, Spacing, and other elements are more touched upon and are far closer to the final product that what is seen in the Low-Fidelity Prototype.
- Full text and items that would be seen in the final product (like fully fleshed out prompts) will be shown in the High-Fidelity Prototype, whereas in the Low-Fidelity Prototype,

they will be summarized more or have placeholders for where different elements such as images will appear.

Our High-Fidelity prototype shows how Mia might interact with the Chat-Bot through clickable buttons, fully fleshed out prompts, and actions that Mia might take.

Figure 15: Home Page

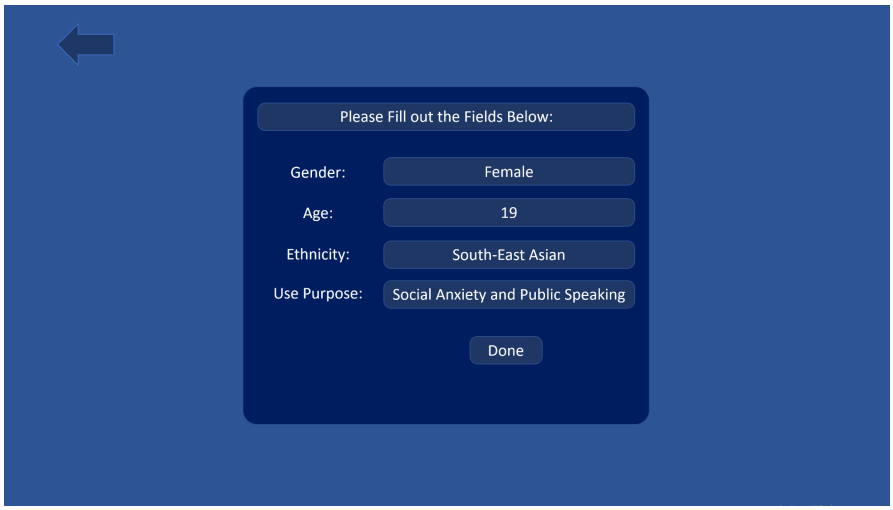


Figure 16: Personalization Page

Personalization Page: When the user opens the prototype, they are greeted with the logo of the Chat Bot and a few different options. Personalization will help the Chatbot tailor the user experience to them.

Signifiers: We made sure that the buttons had their own personal, light-colored backgrounds so that users could differentiate what was a button and what was just plain text. You can see this with the “Go and Chat!”, “Personalize your Experience!”, and “Done” buttons. We added a light blue high to the back button on all pages and made it point left so that users know that they can click it to go back.

Affordances: The two buttons on the homepage let’s users chat with or without personalization. Back arrow lets users go back to the home-page. The done button on the personalization page lets users submit the demographic information.

Design: On the homepage we used centered text alignment and white font to make the text easier to read. We also made the chatbot’s name a larger, serif font while the rest of the text is smaller. We also made the text explaining the buttons smaller since it has more words and we don’t want it to take up too much space. On the personalization page we used a deeper contrast for the backgrounds and we included placeholder text for the demographic information to indicate that the user will be able to fill these out in the final product.

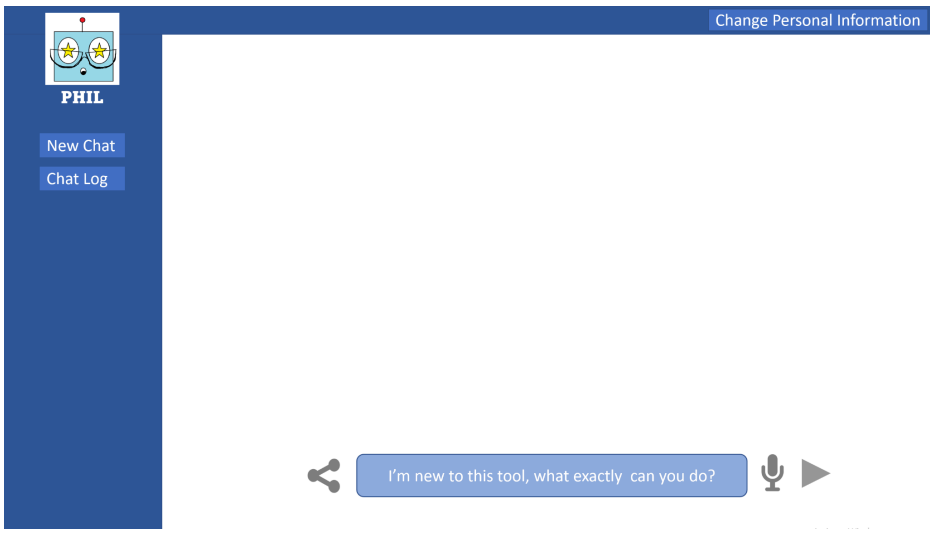


Figure 17: New Chat Page

Chat Pages: This is where the user will start their chat with Phil. It includes a button to submit a question into the bot (the arrow) and some text links for navigation (the settings button and tech support button.

Signifiers: We made sure to make the arrow obvious so that the user feels prompted to click on it. The buttons, again, have their own personal, light-colored backgrounds so that users could differentiate what was a button and what was just plain text. The text links also cause the cursor to turn into a hand, signifying that they are interactable.

Affordances: Lets the user either use the chat bot for its intended purpose (to chat with the AI) or go to other pages (like a new chat, the chat log page, or the report a bug page). The logo can also be clicked on to go back to the home page.

Design: We made sure to keep most of the text links on the left margin so that the main chat body can be free so that the chat doesn't get cramped up once the user starts asking questions and receiving answers. We also used a white font for all of our text to make it more readable. Since this is a prototype for a website, we used san-serif font throughout it (Except for the chatbot's name). The mic button to the right of the text box is just a placeholder since the final product will be able to accept voice dictation.

This is what happens when the arrow button is clicked:

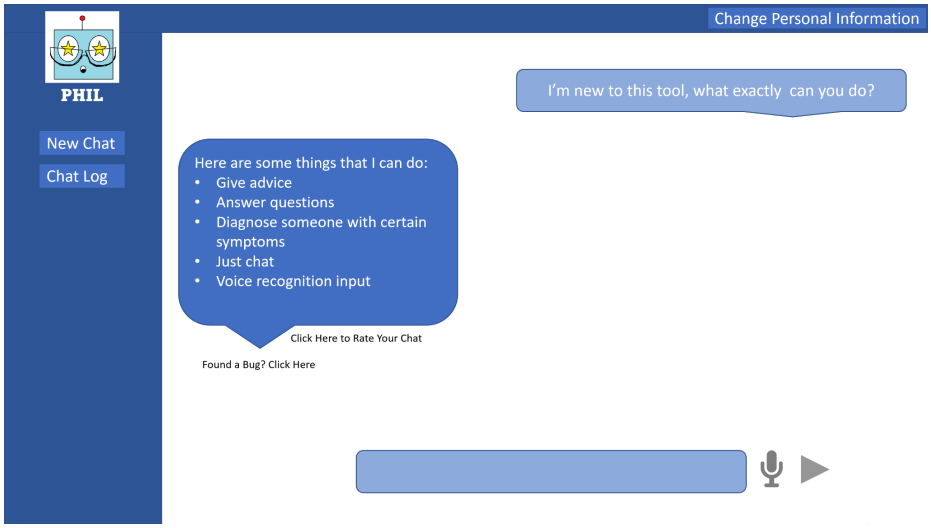


Figure 18: User Help Page

As we can see, the question was sent in and the Chat Bot responded with an answer. What the user can see underneath the response is a prompt that can be clicked on if the user is willing to give a rating for their chat. Now let's go to the Rating Page.

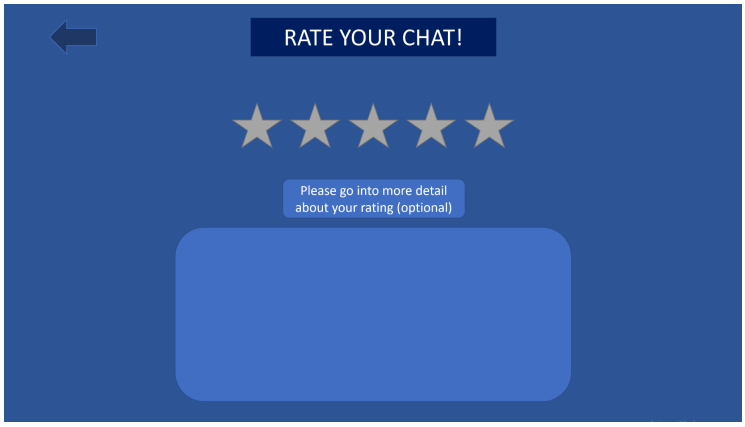


Figure 19: Chat Rating Page

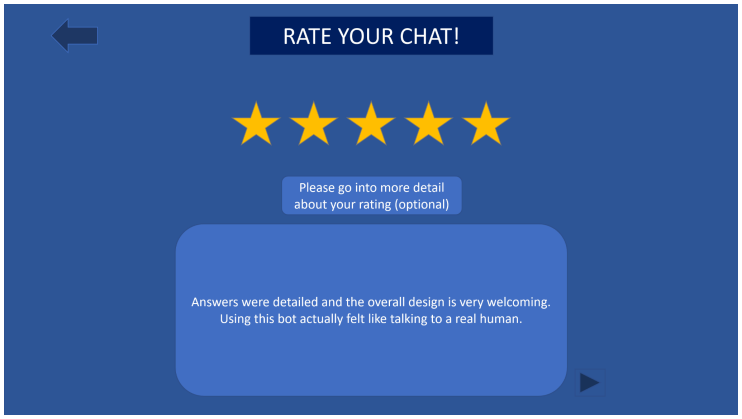


Figure 20: Filled Out Chat Rating Page

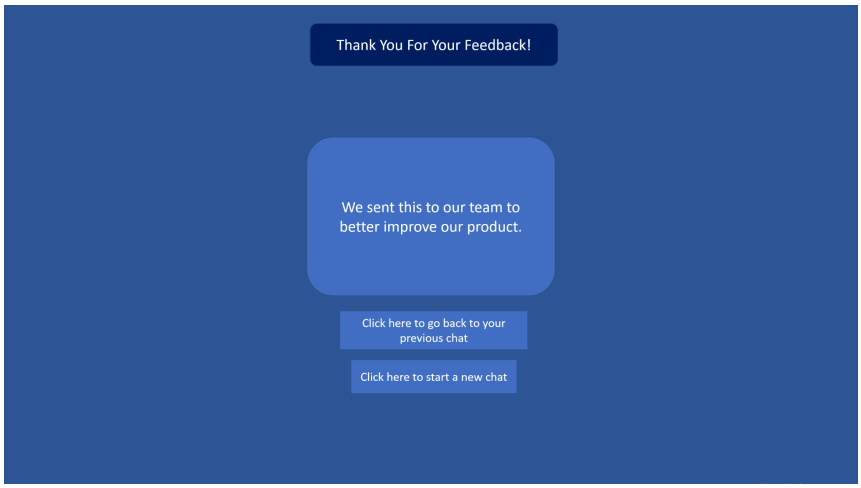


Figure 21: Feedback Confirmation Page

Rating Pages: After clicking on the Rate Your Chat prompt, the user is sent to this page where they can give a star rating.

Signifiers: Stars make the cursor turn into a hand when the user hovers over them and the arrow is made to be obvious since it only appears when the user clicks on a star. We also use the phrase "click here" for the return to chat buttons so that the user knows that they can interact with them.

Affordances: The Stars are actual intractable buttons that can be clicked on to be taken to the page where the user is given the option to give a more detailed review. There is also an arrow button next to the text box that can be clicked on to send in the review and the user will be taken to the Thank You page. On that page the user is given two buttons to choose from. The options will allow them to either go back to their previous chat or to start a new one.

Design: We made sure to follow the same color scheme throughout the prototype with the use of white font and darker colors for background elements. We also made sure to use a bigger font size for headers and smaller font sizes for sub-headers and body text.

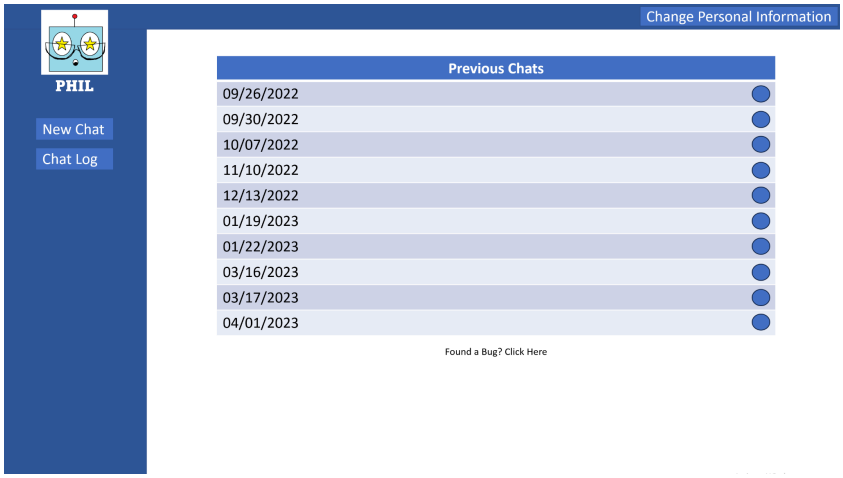


Figure 22: Chat Log Page

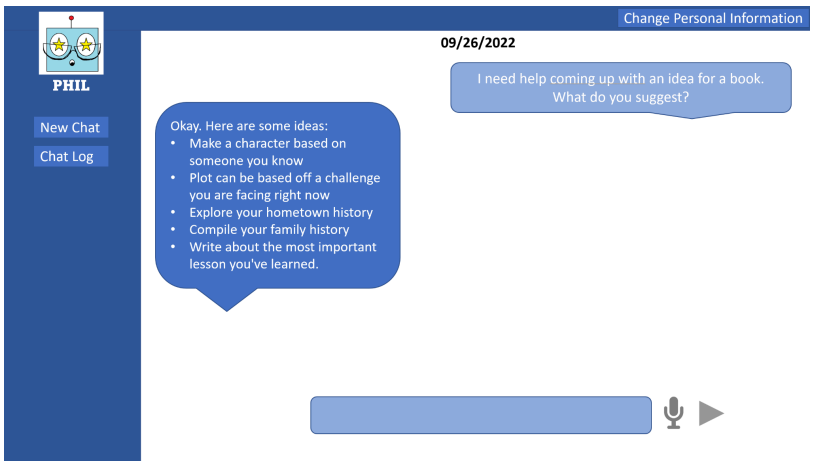


Figure 23: Saved Chat

Chat Log Page: On the chat page, the user can click on the "Chat Log" button to go to this page where they can see all previous chats the user has had.

Signifiers: The text link for "Chat Log" and the chatbot logo will turn the cursor into a hand when the user hovers over them. The buttons, again, have their own personal, light-colored backgrounds so that users could differentiate what was a button and what was just plain text. The "found a bug" button looks better without a background so we just put text telling the user to "click here."

Affordances: The user can click on the date of the most recent chat (the one at the very bottom) or the circle the the right of it to go to a save of that chat from the past. The rest of the dates are

just placeholders. The user can click on the “New Chat” button to go back to a blank chat or click on the “Chat Log” button while they are looking at a save from a previous chat so that they can go back to the Chat Log page. The user can click on “Found a Bug? Click Here” to go to the tech support page.

Design: We are still following the same consistent typography. Alternating colors were used in the chat log to help the user differentiate when a row ends. We also used a darker background for the chatbot’s response like we did previously in figure 17.

Figure 24: Share Page

Share Page: If we look back at figure 17 and figure 22, we can see that there is a share button to the left of the text box. When the user clicks on this, it takes them to the share page and an example is shown above (figure 23).

Signifiers: When hovering over the back button on the share page and the share button on the chat page, the cursor will turn into a hand when the user hovers over them. The back button also has light blue highlighting to make it more obvious that it can be clicked on.

Affordances: The share button takes the user to the share page. The back button takes the user back to their chat that they were previously looking at whether it be a new chat they had or an old one.

Design: We used curved borders for the boxes as we've done throughout the prototype and we used a darker blue as we did before so that the elements within the boxes can be seen better. We used a gray fill for all the buttons that are icons (like the share button) and a dark blue for all arrows. We are also still following the same typography as we have been throughout the prototype, but this time we used blue for the link.

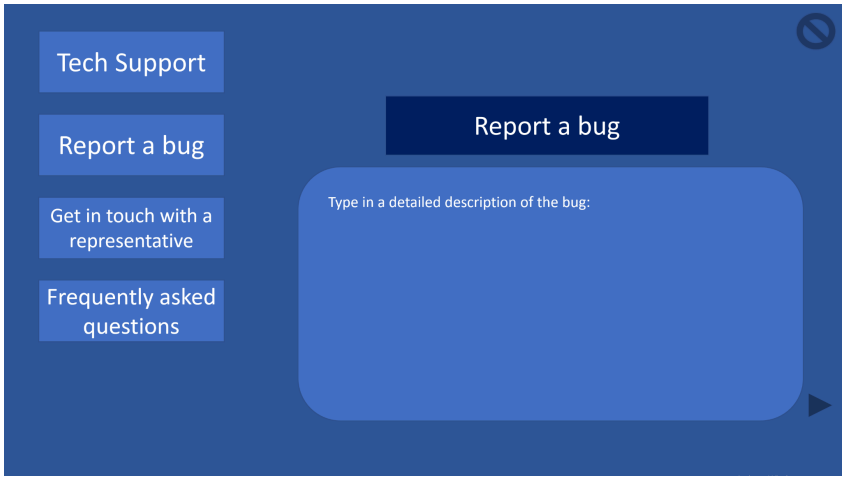


Figure 25: Report a Bug Page

Report a Bug Page: This is where a user of the chatbot would report any problems that they are having with it and where they can also find answers to questions that have been asked before.

Signifiers: This page has a text box with placeholder text telling the user to type. There is a stop button at the top right of this page and an arrow at the bottom right which, when hovered over, turns the cursor into a hand.

Affordances: The user can click the arrow button in the bottom right corner to submit the report (a picture of that page can be seen below. The user can also press the stop button to go back to the chat page. There are placeholder buttons to the left.

Design: The same typography was used here and throughout this prototype with the white, san serif font. Options for different support were kept to the left hand side for organization purposes, like we've been doing throughout the prototype.

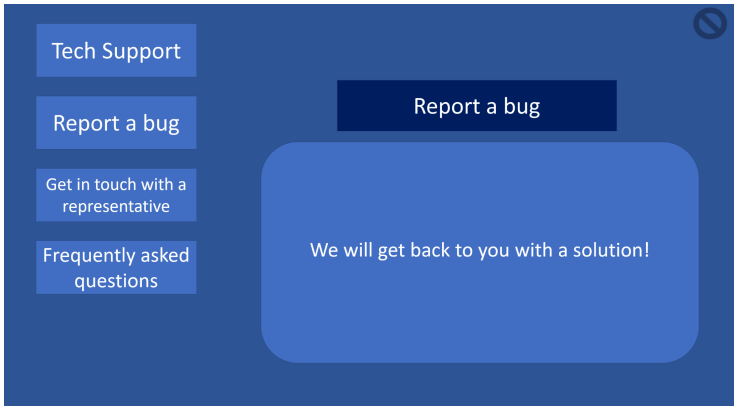


Figure 26: Report Bug Confirmation Page

Evaluation

The user study evaluations were gathered through the Wizard of Oz interviews and experimental usability study. The Wizard of Oz interview aimed to evaluate the current design, while the experimental usability study evaluated the usability of specific tasks. The participants chosen for the studies were people of various ages and backgrounds. We gathered family members, friends, roommates, and neighbors. Though not every person could relate to our personas, we used the resources we had. Plus, having participants of different backgrounds can keep a balanced demographic in our studies.

Wizard of Oz

In the Wizard of Oz study, each group member gathered one or more participants and had them complete one unique task. We informed the participants to speak their thoughts while using the interactive application. Afterwards, we asked them specific questions about the applications.

Unique Tasks:

1. Check log history
2. Rate their experience
3. Create a new chat
4. Reach technical support
5. Share chat link
6. Take a test to personalize the chat

Interview Questions:

1. How was your experience with the current design?
2. During this task, what were some positives or negatives that affected you during the task?
3. While completing the task, were there any visual cues that helped or confused you?
4. Do you have any suggestions to fix or continue helping the user be able to complete the tasks?
5. While navigating the app, did any steps seem easier or harder than others? Please explain.
6. Would you recommend this app to others? Who would you recommend it to?

The overall experience with the app was well received. The participants could all locate the necessary information to navigate the app successfully. The think-aloud strategy helped us immensely discover our design's flaws. For example, a collective joy point was how simple it was to complete the tasks, while a primary pain point was the application's structure. For example, the user was not able to go to the previous page due to there not being any back buttons. To do this, they must go back to the home page and begin the task again. Another pain point was the chatbot's inability to go into advanced topics, even with the personalized test.

The interview questions after the study were leveraged as well. As stated before, the general experience with the current design was good, as it is easy to navigate. The positives of the application were the simplicity and familiarity. The application has a similar format to other chatbot websites, and the layout is consistent. There could be some issues with the application for users who need to become more familiar with navigation in websites and mobile apps. A few areas of improvement are the impracticality of the interface. For example, giving the user an option to stay signed in, even after closing the application, would be helpful. Another example would be allowing users to go to a previous page instead of restarting their task. A significant

pain point in this application is the need for visual queues. Many participants suggested adding text or icons next to the main bar. One participant suggested adding them to headers to signify where the user is in the completion of their task. Most participants stated they would recommend the application to friends and family.

Experimental Usability Study

In the experimental usability study, each group member gathered two participants and had them complete a task in two different ways. The task was to converse with the chatbot through text-based and vocal-based input. We conducted this study to compare the application's usability through typing and voice recognition. We gave them a narrative of what we needed and asked three debriefing questions.

Narrative: We are conducting a study to see the best way for users to interact with our design. Today we will be asking you to type and speak to this chatbot. Please give your honest thoughts and feedback during this study. But first, we have some questions for you:

1. How familiar are you with typing commands?
2. How familiar are you with speaking commands? For example, are you familiar with applications like Alexa and Siri?
3. Do you prefer speaking or typing in applications? Why?

The primary method is typing into the chatbot; the alternative method is speaking to it. We measured how long it took to input information through each technique. We decided to

counterbalance this study by pretesting the experiment's users and dividing them based on their typing and speech speed.

We collected the data through chat logs, notes, and think-aloud methods. We measured the data by timing their typed or speech input per minute and their correctness. We have analyzed the data and placed them in a chart and visual graph below.

Participant #	Time to type (sec)	Time for vocal (sec)
1	60	45
2	53	40
3	57	38
4	48	43
5	56	41
6	49	45
7	47	43
8	50	48
9	56	43
10	64	41
average	53.83	42

Figure 27: Experiment Results Table

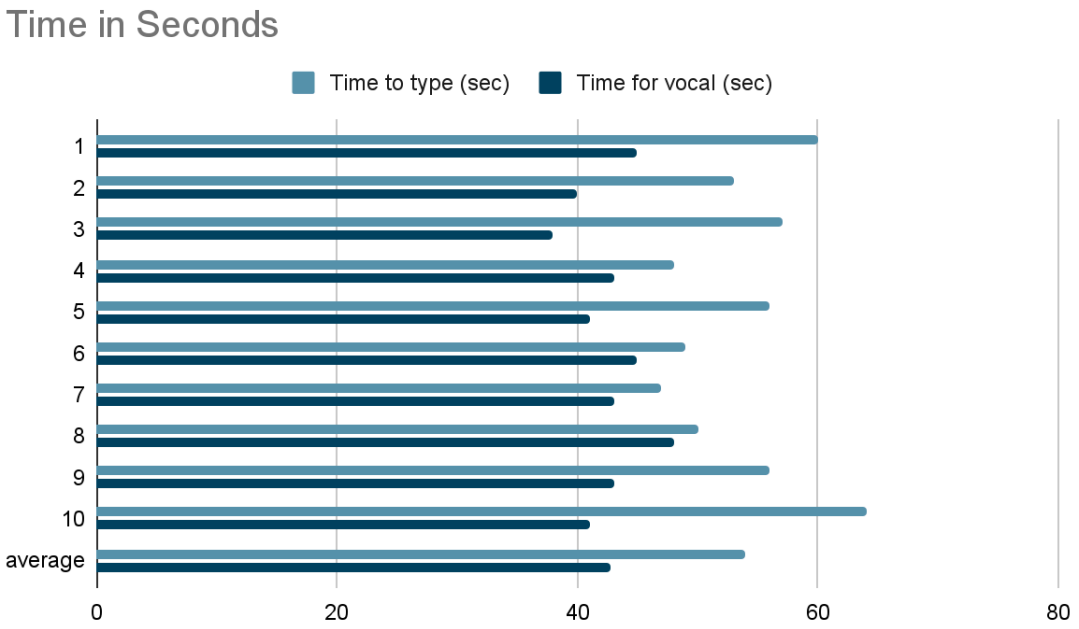


Figure 28: Experiment Results Graph

We took this data to find the best way for users to interact with the interface. The qualitative data helped us see where our goals stood with the potential users, and see what needed to change. The overwhelming theme in our data is that the vocal approach is more efficient than the typed version of this task. This is because most people can speak faster than they can type, which speeds up input time even if the output time stays the same. However, most participants preferred to use the typing version because it was easier to control their input. It is also safer to use in public areas. The testing made it easy to tell which input type was more efficient. Something that was also noted was that the two different methods of input had both pros and cons. Because of this, both plans would likely be implemented into the program. Our high-fidelity prototype was altered to accommodate different types of input.

Modifications

Throughout each study, we altered our design according to the user's needs. The first modification allowed the user to stay signed in despite exiting the application. This way, users won't have to log in whenever they need to use Phil. We decided to add more icons or text to our design to make navigation easier for users. Another way to combat usability issues is by including different input types, such as typing and speaking. Phil must be able to recognize all kinds of voices and languages.

Summary

Our design's primary goals are to make a user-friendly chatbot powered by AI, offer a friendly persona to help someone with anxiety, and be used for general purposes like brainstorming, searching the web, and outputting information. Our Wizard of Oz study and our Experiment usability study targeted the first goal of being user-friendly, specifically in the experiment usability study. We created two ways to perform a similar task to see how efficiently our design would perform. We found that our design was not the fastest way to complete this task, but through our users' feedback, we found that faster is not always better. The experiment usability study indicates we have met our first goal because even though we offered users a quicker way to complete the task, they preferred the one we initially planned. We also found this to be the case in the Wizard of Oz study. The Wizard of Oz study proves our first goal of being user-friendly and provides a friendly persona through this easy-to-use design. While parts of our design, like the friendly persona, depend on the AI and not the design surrounding it, we've proved that with the right AI running in the background, we have most definitely succeeded in our design goals. Something that could be improved is designing a mobile app rather than a desktop app so that the chatbot can be used with easy access. Next, we can implement more personalized features that sync with the user's account, like customized notifications to the user when they haven't logged on in a while. Lastly, we could design a more integrated experience for the user's device by offering widgets for devices that support it or creating a web extension that helps you digest information on a page you're viewing. While a few design features can improve, we are confident that our design goals met our end product successfully.